

JKBOSE 2026

CLASS 11TH

#XUV



CHEMICAL THERMODYNAMICS

LECTURE-3

CHEMISTRY

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ON
TOP



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Topics To Covered

- 1** Work
- 2** Heat
- 3** Heat Capacity

REVISION

TYPES OF PROCESS.

- i) Adiabatic
- ii) Isothermal
- iii) Isochoric
- iv) Isobaric ✓
- v) Cyclic ✓
- vi) Reversible ✓
- vii) Irreversible ✓

State
 &
 path function
 (W, q, C)

WORK.

Irreversible Work

$$W_{\text{irr}} = -P_{\text{ext}} \Delta V$$

Reversible Work

b) REVERSIBLE WORK.

As we know that in a

reversible there are infinite steps, therefore we
proceed

have to calculate the work of each step by integrating.

$$\int dw = - \int_{V_i}^{V_f} P_{ext} dV$$

$$W_{rev} = - \int_{V_i}^{V_f} P_{ext} dV$$

For compression, $P_{ext} > P_{int}$ i.e. $P_{ext} = P_{int} + dp$

For expansion, $P_{ext} < P_{int}$ i.e. $P_{ext} = P_{int} - dp$

$$\text{i.e. } P_{ext} = P_{int} \pm dp$$

$$\text{i.e. } W_{rev} = - \int_{V_i}^{V_f} (P_{int} \pm dp) dV$$

$$\text{or } W_{rev} = - \int_{V_i}^{V_f} P_{int} dV \pm dp dV$$

$dp dV$ is the very small change thus can be neglected.

$$W_{rev} = - \int_{V_i}^{V_f} P_{int} dV$$

To be continued.....

$$W_{rev} = - \int_{V_i}^{V_f} P_{int} dV \rightarrow 1)$$

According to ideal gas equation
 $PV = nRT$

or $P = \frac{nRT}{V}$ Put the value of P
 in equation (1)

$$W_{rev} = - \int_{V_i}^{V_f} \frac{nRT}{V} dV$$

$$\Rightarrow W_{rev} = -nRT \int_{V_i}^{V_f} \frac{dV}{V}$$

$$\Rightarrow W_{rev} = -nRT \int_{V_i}^{V_f} \frac{dV}{V}$$

$$\Rightarrow W_{rev} = -nRT \ln \frac{V_f}{V_i}$$

$$\Rightarrow W_{rev} = -2.303 nRT \log \frac{V_f}{V_i}$$

$$\left\{ \int_{x_1}^{x_2} \frac{dx}{x} = \ln \frac{x_2}{x_1} \right\}$$

$$\left\{ \because \ln = 2.303 \log \right\}$$

REVISION W_{rev}

$$\int dW = - \int_{V_i}^{V_f} P_{ext} dV$$

$$W_{rev} = - \int_{V_i}^{V_f} P_{ext} dV$$

$$P_{ext} = P_{int} \pm dP$$

$$W_{rev} = - \int_{V_i}^{V_f} (P_{int} \pm dP) dV$$

$$= - \int_{V_i}^{V_f} P_{int} dV \pm \cancel{dP dV}$$

$$W_{rev} = - \int_{V_i}^{V_f} P_{int} dV$$

Ideal gas equation

$$PV = nRT \text{ or } P = \frac{nRT}{V}$$

$$W_{rev} = - \int_{V_i}^{V_f} \frac{nRT}{V} dV$$

$$W_{rev} = -nRT \int_{V_i}^{V_f} \frac{dV}{V}$$

$$W_{rev} = -nRT \ln \frac{V_f}{V_i}$$

$$W_{rev} = -2.303 nRT \log \frac{V_f}{V_i}$$

$$\underbrace{W_{rev}}_{\text{reversible}} > \underbrace{W_{irr}}_{\text{irreversible}}$$

SIGN CONVENTION OF WORK.

i) Work done on the system (compression) = +ve

i.e For compression $V_i > V_f$

$$\begin{aligned} \therefore W &= -P_{\text{ext}} (\Delta V) \\ &= -P_{\text{ext}} (V_f - V_i) \\ &= (-ve) (-ve) \\ &= +ve \end{aligned}$$

ii) Work done by the system (expansion) = -ve

For expansion, $V_f > V_i$

$$\begin{aligned} W &= -P_{\text{ext}} \Delta V \\ &= -P_{\text{ext}} (V_f - V_i) \\ &= (-ve) (+ve) \\ &= -ve \end{aligned}$$

Work of free expansion of gas.

Free expansion of gas means expansion of gas vacuum.

$$W = - P_{\text{ext}} \Delta V$$

but $P_{\text{ext}} = 0$

$$W = 0$$

Heat



It is the form of energy, which transfers from one body to another body if there is temperature difference between them.

It is denoted by 'q'.

It is an extensive property & path function.

Unit = Joule.

Heat absorbed by the system = $+q$.
& " released by the system = $-q$.

Heat Capacity

It is defined as amount of heat required to raise the temperature of a system by 1°C or 1K .

It is denoted by 'C'

Mathematically $\rightarrow$

$$C = \frac{q}{\Delta T}$$

It is an extensive property & path function

Unit = J K^{-1} .



SPECIFIC HEAT CAPACITY (C_s)

It is defined as the amount of heat required to the temperature of 1g of substance by 1°C or 1K .

It is denoted by C_s .

Mathematically $\Rightarrow$

$$C_s = \frac{q}{\Delta T \times m} = \frac{C}{m}$$

- It is intensive property & state function.

$$\text{Unit} = \text{J K}^{-1} \text{g}^{-1}$$

MOLAR HEAT CAPACITY (C_m)



It is defined as the amount of heat required to raise the temperature of 1 mole of substance by 1°C or 1K .

It is denoted by C_m .

Mathematically $\Rightarrow$

$$C_m = \frac{q}{\Delta T \times n} = \frac{C}{n}$$

It is intensive property & state function

$$\text{Unit} = \text{J K}^{-1} \text{mol}^{-1}$$

QUESTION



Two litres of an ideal gas at a pressure of 10 atm expands isothermally at 25°C into a vacuum until its total volume is 10 litres. How much heat is absorbed and how much work is done in the expansion?

$$\begin{aligned}\text{Sol:} \rightarrow W &= -P_{\text{ext}} \Delta V \\ &= -0 \times \Delta V \quad (P_{\text{ext}} = 0) \\ &= 0\end{aligned}$$

$$V_i = 2\text{L}$$

$$V_f = 10\text{L}$$

$$P_{\text{ext}} = 10\text{atm}$$

$$T = 25^\circ\text{C}$$

$$= 273 + 25 = 298\text{K}$$

QUESTION



Consider the same expansion, but this time against a constant external pressure of 1 atm. Work = ?

$$\begin{aligned}W_{\text{irr}} &= -P_{\text{ext}} \Delta V \\&= -1 (V_f - V_i) \\&= -(10 - 2) \\&= \underline{-8 \text{ J.}}\end{aligned}$$

Given $\rightarrow$

$$P_{\text{ext}} = 1 \text{ atm.}$$

$$V_f = \underline{10 \text{ L.}}$$

$$V_i = \underline{2 \text{ L.}}$$

QUESTION

$$\log 5 = 0.69$$

$$\begin{aligned} 0^\circ &= 273 \text{ K} \\ 25^\circ &= 273 + 25 \end{aligned}$$



Consider the expansion given in problem 5.2, for 1 mol of an ideal gas conducted reversibly.

Soln $W_{\text{rev}} = -2.303 n \times R \times T \log \frac{V_f}{V_i}$

$$= -2.303 \times \cancel{1 \text{ mol}} \times \cancel{8.314 \text{ J K}^{-1} \text{ mol}^{-1}} \times \cancel{298 \text{ K}} \log \frac{10}{2}$$

$$= -5705.848 \times \log 5$$

$$= -5705.848 \times 0.69 \text{ J}$$

$$= -3937.035 \text{ J}$$

$$n = 1 \text{ mol}$$

$$R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$$

or

$$0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}$$

$$T = 25^\circ \text{C}$$

$$= 273 + 25$$

$$= 298 \text{ K}$$

$$V_f = 10 \text{ L}$$

$$V_i = 2 \text{ L}$$

QUESTION



Calculate the heat required to raise the temperature of 2 moles of a substance with molar heat capacity $C_m = 50 \text{ J/mol K}$ by 10 K.

1 1000 J

2 500 J

3 200 J

4 50 J

$$C_m = \frac{q}{\Delta T \times n}$$

$$\begin{aligned} q &= C_m \times \Delta T \times n \\ &= 50 \text{ J/mol K} \times 10 \text{ K} \times 2 \text{ mol} \\ &= 1000 \text{ J} \end{aligned}$$

$$q = ?$$

$$n = 2$$

$$C_m = 50 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$\Delta T = 10 \text{ K}$$

QUESTION

Heat capacity (C) is defined as:

1 $C = \frac{q}{\Delta T}$

2 $C = q \times \Delta T$

3 $C = \Delta T / q$

4 $C = q$

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Sawal

$$C = \frac{q}{\Delta T}$$

$$C_m = \frac{q}{\Delta T \times n}$$

$$C_s = \frac{q}{\Delta T \times m}$$

QUESTION

COMMUNITY.



Calculate the number of kJ of heat necessary to raise the temperature of 60.0 g of aluminium from 35°C to 55°C. Molar heat capacity of Al is $24 \text{ J mol}^{-1} \text{ K}^{-1}$. Molar mass of Al = 27 g mol^{-1} .

Sol: →

$$C_m = \frac{q}{\Delta T \times n}$$

$$q = C_m \times \Delta T \times n$$

$$= C_m \times (T_2 - T_1) \times \frac{\text{Given mass (60g)}}{\text{Molar mass (27)}}$$

=

J

but

kJ

$$\text{no. of mole} = \frac{\text{Given mass}}{\text{Molar mass}}$$

$$q = ?$$

$$m = 60 \text{ g}$$

$$T_2 = 55^\circ\text{C} = 273 + 55$$

$$T_1 = 35^\circ\text{C} = 273 + 35$$

$$\checkmark C_m = 24 \text{ J mol}^{-1} \text{ K}^{-1}$$

$$\text{Molar mass} = 27 \text{ g/mol.}$$

REVISE KAR LENA.

Thank
You